Chapter 9
Databases

IN THIS CHAPTER YOU WILL:

- learn about the structure and components of a single-table database
- identify the fields necessary for a single-table database
- understand the type of data each of the basic data types represent
- identify appropriate data types for specific data and fields
- describe the purpose and/or need for a primary key in a table
- identify an appropriate primary key for a table
- understand the purpose of SQL scripts
- read and complete SQL scripts that use `SELECT FROM`
- read and complete SQL scripts that use `SELECT FROM WHERE`
- read and complete SQL scripts that use `ORDER BY`
- read and complete SQL scripts that use `SUM`
- read and complete SQL scripts that use `COUNT`.

GETTING STARTED

Your school will have a database of information about you. Find out some of the information the school stores, this could be by talking to your teacher, or other members of staff at the school.

Make sure you consider privacy of data. You can find out the type of information collected, e.g. date of birth, but you cannot collect actual data about specific people or items.

Work in pairs to identify why each item of data is stored by the school. Take it in turns to share your reasons with the rest of the class.

A WORLD OF DATABASES

Almost all organisations, companies, businesses, etc. have to store data. This could be data about the people who work for them, the items they sell, their customers. Almost everything that they need to store is in databases.

Consider a software development company. They have employees who write the program, and then they sell the programs to the public. What will they need to store in a database?

- Information about their employees, e.g. their name and address, the amount they get paid, their bank account details.
- Information about the programs they are developing, e.g. the names, people working on them, the type of program they are, the release data.
- Information about their customers and their orders, e.g. which customer has ordered which program, how much they paid.

The company can then use this database to find information out. They could find out the contact details of all their customers to send them details about new programs. They could find out which employees need to be paid overtime.

Websites also usually have databases, especially if you can log into them, or buy products from them. The website is the front-end, the user interface. When you enter information into the web page, your web browser sends this across the internet to a web server. This web server has a database. It searches the database for the information it needs, then sends it back to your web browser with the HTML code to create the website.

Data is valuable. Companies buy data from other companies so that they can use it for themselves. If you have the names and email addresses of all the people who have bought an item from a clothing company, then you can send them information about your clothing company, possibly even a voucher for 20% off. Then you might gain new customers.



Figure 9.1: A filing cabinet can hold paper databases

CONTINUED

Data can also be stolen. This is an additional threat if the data is stored online, e.g. on a web server. This is because external people can access the web server and access the data on it. This could include personal information such as your bank details. This is why it is important that the web servers have security such as firewalls to attempt to stop this unauthorised access. You can learn more about online security in Chapter 5.

Discussion questions

1 Consider a social media website. What information will they store about their users? What could happen if someone gains unauthorised access to this data?

2 How important is it that a company keeps their data secure? What are the potential consequences of someone gaining unauthorised access to personal data?

9.1 Database structure

Introduction to databases

A **database** is a collection of data that is set up in a structured way. This is usually on a computer, but you could have a paper-based database as long as it is suitably structured.

A database holds data about objects, for example, a person, an item in a shop, an order someone has made, etc. Each of these objects is stored in a **table**.

Each table is then made up of **fields**, a field is an individual piece of data being stored about an object. For a person you could have a field for first name, a field for email address, for phone number, etc.

A collection of these fields about one object is called a **record**. For example, all the information about the person Aarna Singh will be one record.

Most databases have many tables and these are all joined together so that the data is linked. This is outside the scope of the specification. You only need to look at single table databases.

Here's an example database about books in a library. The table is called `books`.

Book name	Author	Publisher	Genre	Fiction
Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
Night stars	K. Mars	Si-fi books	Science fiction	True
Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

This database has five fields; Book name, Author, Publisher, Genre and Fiction.

This database as four records: One for Picking daisies, one for Night starts, one for Dreaming of the sun and one for Cooking for Fun.

KEY WORDS

database: an example of application software to store and manipulate data.

table: a set of data about one type of object, e.g. students.

field: an individual piece of data, e.g. date of birth.

record: all of the records in a table about one object, e.g. all the personal details about one student.

ACTIVITY 9.1

Consider your school. In pairs identify the fields that your school will store about you. Write a list of the fields. Compare your pair's answers with another pair. Were there any differences? Did you add any more to your list?

Data types

Each field has a specific **data type**. This identifies what that field can hold. These are similar to the data types you learnt about in programming (Chapter 8).

Table 9.1 shows the common database data types:

Data type	Description	Examples
Text Alphanumeric	Any combination of letters, symbols and/or numbers. The numbers are treated as text and not numeric, i.e. you can't add them together. These are always speech marks around the data.	"Hello" "2JK8D" "2198" "Hi!!"
Character	A single letter, symbol or number. If it is a number it will be treated as text, i.e. you cannot add to it.	"H" "y" "1" "?"
Boolean	One of two options usually True / False or Yes / No. The data does not have speech marks around it.	True False
Integer	A whole number.	123 999 0 −1928
Real	A number with at least one decimal place.	0.0 1.2 9.99 100.92 −2.9384
Date/time	A date and/or a time.	1/1/2020 8:30 2/3/2020 16:00

Table 9.1: Database data types

KEY WORD

data type: the characteristics of a piece of data. Common data types are string, integer, real and Boolean.

KEY WORDS

Boolean: a data type. True or False.

integer: a data type. Whole numbers.

real: a data type. Decimal numbers.

Primary key

A **primary key** is a unique field in a database. This means that it cannot appear twice in different records. It is used to uniquely identify a record. For example, if you want to find a specific person in a database and you enter their name, there could be dozens of people with the same name. If they had an ID number that is unique to only them, then you can access their record straight away.

KEY WORD

primary key: a unique identifier for a record.

Look at this database again:

Book Name	Author	Publisher	Genre	Fiction
Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
Night stars	K. Mars	Si-fi books	Science fiction	True
Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

Could any of these five fields be unique?

- Book name: there could be two books with the same names.
- Author: one author could write more than one book.
- Publisher: one publisher could publisher more than one book.
- Genre: more than one book could have the same genre.
- Fiction: more than one book could be False, and more than one could be True.

This means that none of these fields is a suitable primary key. So instead we add a key field:

Book ID	BookName	Author	Publisher	Genre	Fiction
1	Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
2	Night stars	K. Mars	Si-fi books	Science fiction	True
3	Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
4	Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

The field Book ID becomes the primary key.

Defining a single-table database

To define a single-table database you will need to decide:

1. the fields that you need
2. the data types for each field
3. the primary key.

Example:

> A shop that sells wool needs to store the items it has in stock. Each item has; a name, e.g. "Sparkle wool", a colour, e.g. "red", a weight, e.g. 2, a price, e.g. $3.99 and a quantity in stock, e.g. 23.

To define a single-table database for the shop, you need to:

1. Identify the fields that you need

Read the description again and for each item it describes select a field name.

Description	Field name
Each item has; a name, e.g. "Sparkle"	Item name
a colour, e.g. "red"	Colour
a weight, e.g. 2	Weight
a price, e.g. $3.99	Price
a quantity in stock, e.g. 23.	Quantity

2. Identify the data types for each field

Look at the example data and the description and select the most appropriate data type from the table.

Field name	Explanation	Data type
ItemName	"Sparkle" – this is letters therefore string or character. There is more than one letter so it is not character.	Text
Colour	"red" this is letters and there is more than 1 character.	Text
Weight	2 – this is numeric, there is no decimal so it is not real.	Integer
Price	$3.99 – this is numeric, there is a decimal so it is not integer.	Real
Quantity	23 – this is numeric, there is no decimal so it is not real.	Integer

3. Identify a primary key

None of these fields are unique, you could have two with the same name, same colour, etc. Therefore a new ID field needs adding to become the primary key.

Field name	Data type
IDNumber	Text
ItemName	Text
Colour	Text
Weight	Integer
Price	Real
Quantity	Integer

COMPUTER SCIENCE IN CONTEXT

Every time someone creates a new database they have to go through the same process you have just been through. Consider a library. When they set up a new database they have to program the database with the fields and data types that they need to use.

PROGRAMMING TASK 9.1

A clothes store needs a single-table database to store data about its stock. It needs to store the barcode of each item, the name of the product, the colour, size, quantity in stock and whether an order for more products has been made or not.

Getting started

1 Explain what is meant by a record and a field in a database.

2 Identify **four** data types and describe the data they store.

3 Identify what is meant by a primary key.

Practice

1 Identify the fields for your database table.

2 Identify the data types for your fields.

3 Identify the primary key for your table.

4 Create a paper-based table for this system and complete several records.

Challenge

This task goes beyond the range of the specification.

Open a database software, e.g. Microsoft Access, and find out how to create a table and add data to it. Create the database for the shop.

Questions

1 Complete the table by giving an example of each type of data.

Data type	Example
Boolean	
Integer	
Text	
Character	
Real	
Date/time	

2 Describe the format of a database table.

3 Identify the purpose of a primary key.

4 a A social media website stores the following information about its users: username (e.g. daisy123), password (e.g. HWHRU2D), online name (e.g. Daisy), date account created (e.g. 9/9/2019), number of friends (e.g. 38).
Identify the fields and data types for a single-table database for the website.

b Identify an appropriate primary key for the table.

9.2 SQL

SQL stands for Structured Query Language. This is a standard language that is used across most types or database to perform actions such as:

- define tables
- change tables
- add data to tables
- search for data from tables (query)
- perform calculations using data from tables.

In this specification you only need to know the last two of these: search for data and perform calculations using data.

When you write some SQL code you are writing an SQL **script.** This is series of statements that are commands that are then executed and one or more values are returned.

KEY WORDS

Structured Query Language (SQL): a standard language used to define and manipulate databases.

script: a set of statements that are executed.

9.3 SELECT ... FROM

This allows you select a set of fields from a table. All of the data in each field will be returned.

This table is called Books.

Book ID	BookName	Author	Publisher	Genre	Fiction
1	Picking daisies	J. Frank	Cambridge horticulture	Gardening	False
2	Night stars	K. Mars	Si-fi books	Science fiction	True
3	Dreaming of the sun	P. Yu	Si-fi books	Science fiction	True
4	Cooking for fun	W. Crisp	Cookery penguin	Cookery	False

Select has the format:

```
SELECT field
FROM table
```

Single field:

```
SELECT BookName
FROM Books
```

This query will return:

Picking daisies
Night stars
Dreaming of the sun
Cooking for fun

IMPORTANT: The field names and table name must be exact. If the field name in the table is 'Genre' then putting 'genre' in the query is incorrect. It is case sensitive.

Multiple fields:

```
SELECT BookName, Genre
FROM Books
```

Notice how there are no commas or any other symbol between each field.

This will return

Picking daisies	Gardening
Night stars	Science fiction
Dreaming of the sun	Science fiction
Cooking for fun	Cookery

The order the fields appear in the query are important. The data will be returned in the order they appear in the query.

Example:

```
SELECT BookName, Author
FROM Books
```

will return book name then author:

Picking daisies	J. Frank
Night stars	K. Mars
Dreaming of the sun	P. Yu
Cooking for fun	W. Crisp

The command SELECT is followed by the field name.

The command FROM is followed by the name of the table.

The case, spellings and order are all important in the answer. Putting commas after each one is incorrect. Putting them one after another on a line is incorrect. Each record is on a new line.

Each field is separated by a comma (,).

Each record is on a new line, Picking daisies and Gardening are on the same line because they are part of the same record.

whereas:

```
SELECT Author, BookName
FROM Books
```

will return author then book name:

J. Frank	Picking daisies
K. Mars	Night stars
P. Yu	Dreaming of the sun
W. Crisp	Cooking for fun

COMPUTER SCIENCE IN CONTEXT

SQL is known as a standard. This means that it is used (almost) universally, no matter what software you are using, or type of database. This means that anyone who knows SQL can use any type of database simply by entering the same SQL code.

Questions

These questions use the following database table, PRODUCTS, about products for sale in a shop.

ID number	Name	Type	Cost	Quantity in stock
12GH	Saturn	Chocolate	0.56	100
59RF	Vanilla	Fudge	1.99	20
3WR	Peanuts	Nuts	2.56	2
52D	Jazz	Fruit	0.23	33
6SE	Signet	Fudge	1.48	5

5 Write the results returned by the following SQL script:

```
SELECT ID number
FROM PRODUCTS
```

6 Write the results returned by the following SQL script:

```
SELECT Name, Type
FROM PRODUCTS
```

7 Complete the SQL script to display the prices of all the products.

```
SELECT .....
FROM PRODUCTS
```

8 Complete the SQL script to display the ID number and name of all the products.

```
..... ID Number, .....
FROM PRODUCTS
```

9 Complete the SQL scripts to display the Name, Cost and Quantity in stock for all items.

```
..... ....., ....., Quantity in stock
..... PRODUCTS
```

9.4 SELECT ... FROM ... WHERE

Including a WHERE clause in the SELECT statement allows you to select only specific data. For example, you only want gardening books, or items that cost more than $3.00.

Single condition (single WHERE)

This has only one condition. The condition needs a **logical operator**. These are the same operators that you use in programming (Table 9.2).

KEY WORD

logical operator: a symbol that performs a comparison resulting in True or False. Can be equals, not equal to, less than, less than or equal to, greater than, greater than or equal to.

Operator	Description	Example WHERE
=	Equal to.	WHERE Fiction = True
<	Less than.	WHERE Cost < 3.99
<=	Less than or equal to.	WHERE Cost <= 10
>	Greater than.	WHERE Quantity > 0
>=	Greater than and equal to.	WHERE Quantity >= 100
<>	Not equal to.	WHERE Cost <> 4.0

Table 9.2: Logical operators

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

Select where has the format:

```
SELECT SUM field
FROM table
WHERE condition
```

Equal to

```
SELECT FirstName, LastName, Posted
FROM Orders
WHERE Posted = Yes
```

This will return:

Simon	Tong	Yes
Mei	Liu	Yes

Not equal to

```
SELECT Order ID, Posted
FROM Orders
WHERE Posted <> Yes
```

This query returns:

1A	No
1D	No

Less than

```
SELECT Order ID
FROM Orders
WHERE NumberItems < 10
```

This query returns:

1B
1C

Less than or equal to

```
SELECT NumberItems, TotalCost
FROM Orders
WHERE TotalCost <= 6.00
```

This query returns:

2	3.99
1	6.00

Greater than

```
SELECT FirstName, LastName
FROM Orders
WHERE NumberItems > 20
```

This query returns:

Jenny	Silver

Greater than or equal to

```
SELECT Posted, FirstName
WHERE TotalCost >= 3.99
```

This query returns:

Qirat	No
Simon	Yes
Mei	Yes
Jenny	No

Questions

These questions use the following database table, MARKS, about students and their marks in tests.

Student ID	Subject	Test name	Percentage	Grade
123	Science	Physics 1	10	Fail
596	Computer Science	Programming 2	53.3	Pass
123	Maths	Trigonometry 6	10	Pass
023	Maths	Trigonometry 6	50	Pass
802	Science	Physics 1	25	Fail
806	Computer Science	Programming 2	46.6	Pass
023	Maths	Trigonometry 7	81	Merit
596	Science	Physics 1	50	Pass

10 Write the results returned by the following SQL script:

```
SELECT Student ID, Subject
FROM MARKS
WHERE Subject = "Programming 2"
```

11 Write the results returned by the following SQL script:

```
SELECT Student ID, Subject, Grade
FROM MARKS
WHERE Grade = "Pass"
```

12 Complete the SQL script to display the Test name, percentage and grade where the percentage is more than 50.

```
..... Test name, Percentage, .....
FROM .....
WHERE Percentage ..... .....
```

13 Complete the SQL script to display the student ID, subject and percentage where the subject is Science.

```
SELECT Student ID, ....., .....
..... MARKS
..... SUBJECT = .....
```

14 Complete the SQL script to display the student ID, test name and percentage where the percentage is less than or equal to 25.

```
..... ....., Test name, Percentage
..... .....
WHERE ..... <= .....
```

More than one condition

Combining conditions requires a **Boolean operator** between the conditions. These are similar to those that you used in programming (Table 9.3).

KEY WORD

Boolean operator: a symbol that joins multiple logical comparisons, can be AND OR or NOT.

Operator	Description	Example WHERE
AND	Only data that meets both conditions will be returned.	WHERE Fiction = TRUE AND Cost < 3.99. Only records where Fiction is TRUE and Cost is less than 3.99 will be displayed.
OR	Data that meets one condition, or both conditions will be returned.	WHERE Fiction = TRUE OR Cost < 3.99. Any records where Fiction is TRUE will be returned. Any records where Cost < 3.99 will be returned. If a record meets both then this will be returned only once.

Table 9.3: Boolean operators

Multiple conditions have the format:

```
SELECT field
FROM table
WHERE condition1 logical operator condition2
```

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

AND

```
SELECT Order ID, NumberItems, Posted
FROM Orders
WHERE NumberItems > 1 AND Posted = Yes
```

This query returns:

1B	2	Yes

OR

```
SELECT FirstName, TotalCost, Posted
FROM Orders
WHERE TotalCost < 5 OR Posted = No
```

This query returns:

Qirat	21.99	No
Simon	3.99	Yes
Jenny	59.65	No

Selecting values in, or not in, a condition

The fields in the SELECT do not need to include those in the WHERE.

Example:

```
SELECT FirstName, LastName
FROM Orders
WHERE Posted = Yes
```

This will return the first name and last name of orders where posted is true. It does not return the field posted. It will return:

Simon	Tong
Mei	Liu

COMPUTER SCIENCE IN CONTEXT

If you log into a website then the code behind the scenes runs a query to find out what your username and password are, and then compares what you entered to these details. This is likely to use an SQL SELECT ... FROM ... WHERE statement.

Questions

These questions use the following database table, `WEATHER`, that store weather readings.

Date	Time	Temperature	Wind speed	Humidity
11/12/2019	10:00	20.3	22	59
11/12/2019	13:00	23.7	15	63
11/12/2019	15:00	24.0	13	61
11/12/2019	18:00	23.9	16	60
12/12/2019	15:00	18.4	20	52
12/12/2019	18:00	17.2	18	48

15 Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
WHERE Date = 11/12/2019 AND time < 15:00
```

16 Write the results returned by the following SQL script:

```
SELECT Date, Time, Humidity
FROM WEATHER
WHERE Temperature > 20 AND Wind speed < 20
```

17 Write the results returned by the following SQL script:

```
SELECT Date, Time
FROM WEATHER
WHERE Temperature > 20 OR Wind speed >= 20
```

18 Complete the SQL script to display the time when the date is 12/12/2019 and the humidity is more than 50.

```
SELECT .....

FROM .....

WHERE .....= 12/12/2019 ..... Humidity > 50
```

19 Complete the SQL script to wind speed and temperature, when the wind speed is either below 15 or above 20.

```
..... Wind speed, .....

FROM WEATHER

WHERE Wind speed ..... 15 ..... Wind speed ..... 20
```

PROGRAMMING TASK 9.2

These questions will need your shop database table from Programming Task 9.1.

Getting started

1 Identify the command words in an SQL search script.

2 Identify *four* logical operators and describe the function of each.

3 Identify *two* Boolean operators and describe the function of each.

Practice

1a Complete the SQL script to return the name of all the products in the store.

```
SELECT .....

FROM .....
```

1b Identify the values that will be returned from your query in **1a**.

2a Complete the SQL script to return the name of all products where an order for more products have been made.

```
SELECT .....

FROM .....

WHERE ..... ..... .....
```

2b Identify the values that will be returned from your query in 2a.

3a Complete the SQL script to return the product barcode, colour and size where the quantity in stock is less than 10 and an order for more products has not been made.

```
SELECT .....,  .....,  .....

FROM .....

WHERE ..... ..... .....

AND ..... ..... .....
```

3b Identify the values that will be returned from your query in **3a**.

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Challenge

This task goes beyond the range of the specification.

Find out how to write SQL queries in the database software you have used. Write the queries for practice and run them in your database.

9.5 Order by

This command allows you to state which order the result will be displayed in. There are two options, ascending (`ASC`) and descending (`DESC`). The default is ascending.

The order by statement comes after the `SELECT ... FROM` or `SELECT ... FROM ... WHERE`.

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

Example 1:

```
SELECT OrderID, NumberItems
FROM Orders
ORDER BY NumberItems ASC
```

This query will return:

1C	1
1B	2
1A	20
1D	29

Example 2:

```
SELECT OrderID, TotalCost, Posted
FROM Orders
WHERE Posted = Yes
ORDER BY TotalCost DESC
```

This query will return:

1C	6.00	Yes
1B	3.99	Yes

Questions

These questions use the following database table, WEATHER, that store weather readings.

Date	Time	Temperature	Wind speed	Humidity
11/12/2019	10:00	20.3	22	59
11/12/2019	13:00	23.7	15	63
11/12/2019	15:00	24.0	13	61
11/12/2019	18:00	23.9	16	60
12/12/2019	15:00	18.4	20	52
12/12/2019	18:00	17.2	18	48

20 Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
ORDER BY Temperature
```

21 Write the results returned by the following SQL script:

```
SELECT Date, Time, Temperature
FROM WEATHER
ORDER BY Humidity DESC
```

22 Complete the SQL script to display the date, wind speed and humidity in ascending order of wind speed.

```
..... Date, Wind speed, Humidity
..... WEATHER
..... Wind speed
```

23 Complete the SQL script to display the date, time and temperature, in descending order of humidity.

```
SELECT Date, Time, .....
FROM WEATHER
ORDER BY ..... .....
```

24 Complete the SQL script to display the temperature, wind speed and humidity in descending order of temperature.

```
..... ....., Wind speed, .....
FROM .....
..... ..... .....
```

9.6 SUM

SUM command will add up the field that is written after it. It is written in the select statement in the format:

```
SELECT SUM(field)
FROM table
```

You can also use `WHERE` conditions. e.g.

```
SELECT SUM(field)
FROM table
WHERE condition
```

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

Example 1:

```
SELECT SUM(NumberItems)
FROM Orders
```

This query will return:

52

It is calculated from adding together all of the number of items (20 + 2 + 1 + 29).

Example 2:

```
SELECT SUM(TotalCost)
FROM Orders
WHERE Posted = No
```

This query will return:

81.64

It is calculated from adding together all the total costs where posted is No (21.99 + 59.65).

9.7 COUNT

This will count how many records meet the criteria, or how many fields are in the table.

Count is in the format:

```
SELECT COUNT(field)
FROM table
```

The following queries will use the following table, `Orders`:

Order ID	FirstName	LastName	NumberItems	TotalCost	Posted
1A	Qirat	Raja	20	21.99	No
1B	Simon	Tong	2	3.99	Yes
1C	Mei	Liu	1	6.00	Yes
1D	Jenny	Silver	29	59.65	No

Example 1:

```
SELECT COUNT(OrderID)
FROM table
```

The query will return:

4

Example 2:

```
SELECT COUNT(NumberItems)
FROM table
WHERE Posted = Yes
```

The query will return:

2

Example 3:

```
SELECT COUNT(OrderID)
FROM Orders
WHERE NumberItems > 10
```

The query will return:

2

SKILLS FOCUS 9.1

SUM AND COUNT

SUM and COUNT are both mathematical operations that perform different functions.

SUM

This performs addition of a group of values.

Example 1, a set of data.

1 + 3 + 8 + 9 + 3 = 24

Example 2, a table of data.

10
2
9
30
4
5

The SUM is 10 + 2 + 9 + 30 + 4 + 5 = 60

Example 3, an SQL query.

This query will use the following table, SALES.

CONTINUED

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	2	3.00
5	4	11.50

```
SELECT SUM(Total)
FROM SALES
```

The command word `SUM` tells you it needs you to add values together. The field in the brackets, `Total`, tells you to add together the Total column.

20.00 + 16.25 + 8.99 + 3.00 + 11.50 = 59.74

When you are answering questions such as what does this script return when `SUM` is being used, the only answer is the number. In Example 3, the answer to the question would just be: 59.74

Example 4, table with criteria using the table `SALES`.

```
SELECT SUM(Total)
FROM SALES
WHERE Customer ID = "2"
```

This time there is a criteria. Only records with Customer ID being 2 are needed.

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	2	3.00
5	4	11.50

There are two records with Customer ID 2. The total for these two records are added together.

16.25 + 3.00 = 19.25

The SQL script will return:

19.25

COUNT

This counts the number of records (rows). This is not related to the values in the fields, just how many records there are that meet the criteria.

Example 1, a set of data.

1, 3, 8, 9, 3. There are **5** values.

CONTINUED

Example 2, a table of data.

10
2
9
30
4
5

There are **6** values

Example 3, an SQL query.

This query will use the following table, SALES.

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	3	3.00
5	4	11.50

```
SELECT COUNT(Total)
FROM SALES
```

The command word COUNT tells you that you are counting records. The field in the brackets, Total, tells you which to count.

20.00, 16.25, 8.99, 3.00, 11.50

The value returned will be: 5

When you are answering questions such as what does this script return when COUNT is being used, the only answer is the number. In Example 3, the answer to the question would just be: 5

Example 4, table with criteria using the table SALES.

```
SELECT COUNT(Total)
FROM SALES
WHERE Customer ID = "2"
```

This time there is a criteria. Only records with Customer ID being 2 are needed.

Order ID	Customer ID	Total
1	6	20.00
2	2	16.25
3	8	8.99
4	2	3.00
5	4	11.50

CONTINUED

Count how many records have Customer ID 2. There are 2 records.

The SQL script will return:

2

Questions

These will use the following table, PRODUCTS:

ID number	Name	Type	Cost	Quantity in stock
12GH	Saturn	Chocolate	0.56	100
59RF	Vanilla	Fudge	1.99	20
3WR	Peanuts	Nuts	2.56	2
52D	Jazz	Fruit	0.23	33
6SE	Signet	Fudge	1.48	5

1 What is the difference between SUM and COUNT?

2 What will be returned when the following SQL script is run?

```
SELECT SUM(Quantity in stock)
FROM PRODUCTS
```

3 What will be returned when the following SQL script is run?

```
SELECT COUNT(ID number)
FROM PRODUCTS
```

4 What will be returned when the following SQL script is run?

```
SELECT SUM(Cost)
FROM PRODUCTS
WHERE Quantity in stock > 10
```

5 What will be returned when the following SQL script is run?

```
SELECT COUNT(Name)
FROM PRODUCTS
WHERE Cost < 2.00
```

ACTIVITY 9.2

There are other mathematical functions that SQL can perform. Find at least one other example of a mathematical function and how to write an SQL script using it.

Take one of the example databases and write a script using your function.

Peer Assessment

Swap your new function and your script with a partner. Research the new function your partner has used and check whether their script is accurate. If not, discuss the error and how to solve it.

PROGRAMMING TASK 9.3

These questions will need your shop database table from Programming Task 9.1.

Getting started

1 Identify the commands to order a query in descending order.

2 Identify the commands to add together a set of values.

3 Identify the commands to count how many records there are.

Practice

1a Complete the SQL script to return the name of all the products in the store in descending order of quantity in stock.

```
SELECT .....
FROM .....
ORDER BY .....
```

1b Identify the values that will be returned from your query in **1a**.

2a Complete the SQL script to return the total cost of all products.

```
SELECT ..... (.....)
FROM .....
```

2b Identify the values that will be returned from your query in **2a**.

3a Complete the SQL script to return how many items have been ordered for more products.

```
SELECT ..... (.....)
FROM .....
WHERE ..... ..... .....
```

3b Identify the values that will be returned from your query in 3a.

4 Complete the SQL script to return how many items are red, and have more than 100 in stock, in ascending order by the number in stock.

```
SELECT ..... (.....)
FROM .....
WHERE ..... ..... .....
AND ..... ..... .....
..... .....
```

Challenge

This task goes beyond the range of the specification.

Write each of the queries in the practice section in your database and execute them to see the results.

REFLECTION

How did you approach Programming Task 9.3? Did you have other queries with you to check your syntax and to help you answer the questions? How will you make sure you remember the commands?

Questions

These questions use the following database table, ORDERS, that stores details about orders from a shop.

Order ID	FirstName	LastName	NumberItems	TotalCost
15	Keanu	Kawai	5	20.59
16	Keanu	Kawai	10	101.12
17	Eka	Loke	3	30.50
18	Keanu	Kawai	2	14.00
19	Ariana	Wang	4	21.10
20	James	Smith	6	44.60
21	James	Smith	20	156.90
22	Ajay	Gupta	1	10.00

25 Write the results returned by the following SQL script:

```
SELECT SUM(NumberItems)
FROM ORDERS
```

26 Write the results returned by the following SQL script:

```
SELECT COUNT(NumberItems)
FROM ORDERS
```

27 Write the results returned by the following SQL script:

```
SELECT COUNT(NumberItems)
FROM ORDERS
WHERE NumberItems > 10
```

28 Write the results returned by the following SQL script:

```
SELECT SUM(TotalCost)
FROM ORDERS
WHERE FirstName = "Keanu"
```

29 Complete the SQL script to display how many orders James Smith has made.

```
SELECT ..... (FirstName)

FROM ORDERS

WHERE FirstName = "James" ..... LastName =
"Smith"
```

30 Complete the SQL script to calculate the total cost of orders where 5 or more items have been ordered.

```
SELECT ..... (TotalCost)

FROM ORDERS

..... NumberItems ..... 5
```

31 Complete the SQL script to calculate how many orders cost more than 50.00

```
SELECT ..... (NumberItems)

FROM .....

WHERE ..... ..... .....
```

SUMMARY

A database stores data about objects such as items, people, orders.
A field is one piece of data in a table.
A record is a set of fields about one object.
Each field has a data type.
The text or alphanumeric data type stores any combination of letters, numbers and symbols.
The character data type stores one letter, number or symbol.
The Boolean data type stores one of two values, either True or False.
The Integer data type stores whole numbers.
The real data type stores decimal numbers.
The date/time data type can store a date, time, or a date and time.
A primary key is a unique identifier in a table.
A primary key is used to uniquely identify each record.
SQL stands for Structured Query Language and is a standard for querying databases.
`SELECT ... FROM` returns all the data in the field given.
`SELECT ... FROM ... WHERE` returns all the data in the field where the `WHERE` condition is true.
`ORDER BY` can be `ASC` (ascending) or `DESC` (descending) and will return the data in the order specific by the field given.
`SUM` will total the values in the given field.
`COUNT` will return how many records meet the criteria.

EXAM-STYLE QUESTIONS

1 **Define** the terms record and field. Give an example of each. **[2]**

2 **Describe** the following data types and give an example for each.

- Boolean
- Real
- Date / time
- Text **[4]**

3 A cinema stores data about the films it is showing. It stores the screen number (for example, 1), the film (for example, The Dark), the date and time of the showing, the number of seats sold (e.g. 150).

Define the single-table database for the cinema by identifying the fields, data types, and primary key for the table. **[7]**

COMMAND WORDS

define: give precise meaning.

describe: state the points of a topic / give characteristics and main features.

CONTINUED

4 A hairdressers uses a database, APPOINTMENTS, to store upcoming appointments. Part of this database is shown:

Appointment ID	Date	Time	Employee ID	Customer first name	Customer last name	Appointment type
12	1/2/2020	14:00	JP	James	Smith	Cut
13	1/2/2020	14:30	AD	Ajay	Gupta	Cut and dry
14	1/2/2020	16:00	AD	Eka	Loke	Colour
15	1/2/2020	9:45	JP	Keanu	Kawai	Cut and dry
16	2/2/2020	11:15	JP	Ariana	Wang	Cut
17	2/2/2020	12:30	AD	Osandi	Bandara	Colour

a **Identify** how many records are in the table. **[1]**

b Identify how many fields are in the table. **[1]**

c **State** what is meant by a primary key, ***and*** identify the primary key in the table APPOINTMENTS. **[2]**

d Identify the output from this SQL script:

```
SELECT EmployeeID, Appointment type
FROM APPOINTMENTS
```

[2]

e Identify the output from this SQL script:

```
SELECT COUNT(AppointmentID)
FROM APPOINTMENTS
WHERE Appointment type = "Cut"
```

[1]

f Complete the SQL script to return the first and last name of all customers with appointments on 1/2/2020.

```
..... Customer first name, Customer last name
FROM APPOINTMENTS
WHERE ..... ..... .....
```

[4]

g Complete the SQL script to return the number of appointments JP has on 1/2/2020.

```
SELECT ..... (Employee ID)
..... APPOINTMENTS
..... Employee ID ..... .....
```

[5]

[Total: 16]

COMMAND WORDS

identify: name / select / recognise.

state: express in clear terms.

CONTINUED

5 A single-table database, SALES, stores details of people who have bought books online.

Sales ID	Book ID	Date	Customer ID	Quantity bought	Total Cost
5	10	3/3/2020	1GHF2	1	6.99
6	151	3/3/2020	34FDD	2	10.00
7	230	3/3/2020	34FDD	1	12.00
8	230	3/3/2020	1GHF2	1	12.00
9	10	3/3/2020	P0OO1	2	13.98
10	88	4/4/2020	15DTB	2	10.50
11	209	4/4/2020	1GHF2	3	12.00
12	151	4/4/2020	34FDD	5	25.00
13	28	4/4/2020	34FDD	2	20.00

a Identify the most appropriate data types for the following fields:

- Sales ID
- Date
- Customer ID
- Quantity bought
- Total Cost

[5]

b **Give** the result from the following SQL script:

```
SELECT Date, CustomerID
FROM SALES
WHERE CustomerID = "1GHF2"
```

[2]

c Complete the following SQL script to return the customer ID and total cost of each order where more than 1 book were ordered.

```
SELECT Customer ID, .....
FROM .....
WHERE Total Cost ..... .....
```

[4]

d Give the result from the following SQL script.

```
SELECT Sales ID, BookID
FROM SALES
WHERE Date = 4 / 4/ 2020 AND Total Cost <= 15
```

[2]

COMMAND WORD

give: produce an answer from a given source or recall / memory.

CONTINUED

e Complete the following SQL script to calculate the total number of books sold on 4/4/2020.

```
SELECT ..... (Quantity bought)
..... SALES
WHERE ..... = .....
```

[4]

f Complete the following SQL script to calculate the total cost of all the orders.

```
SELECT ..... (Total Cost)
..... .....
```

[3]

[Total: 20]

SELF-EVALUATION CHECKLIST

After studying this chapter, think about how confident you are with the different topics. This will help you to see any gaps in your knowledge and help you to learn more effectively.

You might find it helpful to rate how confident you are for each of these statements when you are revising. You should revisit any topics that you rated 'Needs more work' or 'Getting there'.

I can...	See topic	Needs more work	Getting there	Confident to move on
learn about the structure and components of a single-table database.	9.1			
identify the fields necessary for a single-table database.	9.1			
identify appropriate data types for specific data and fields.	9.1			
identify an appropriate primary key for a table.	9.1			
understand the purpose of SQL scripts.	9.2			
read and complete SQL scripts that use `SELECT FROM`	9.3			
read and complete SQL scripts that use `SELECT FROM WHERE`.	9.4			
read and complete SQL scripts that use `ORDER BY`.	9.5			
read and complete SQL scripts that use `SUM`.	9.6			
read and complete SQL scripts that use `COUNT`.	9.7			